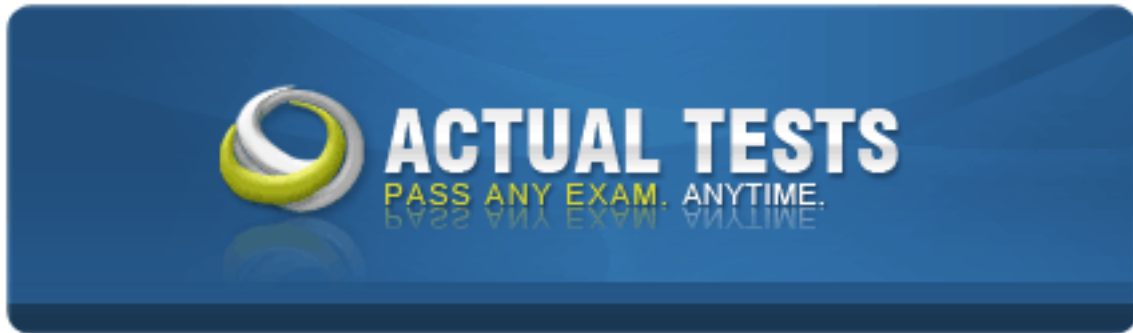


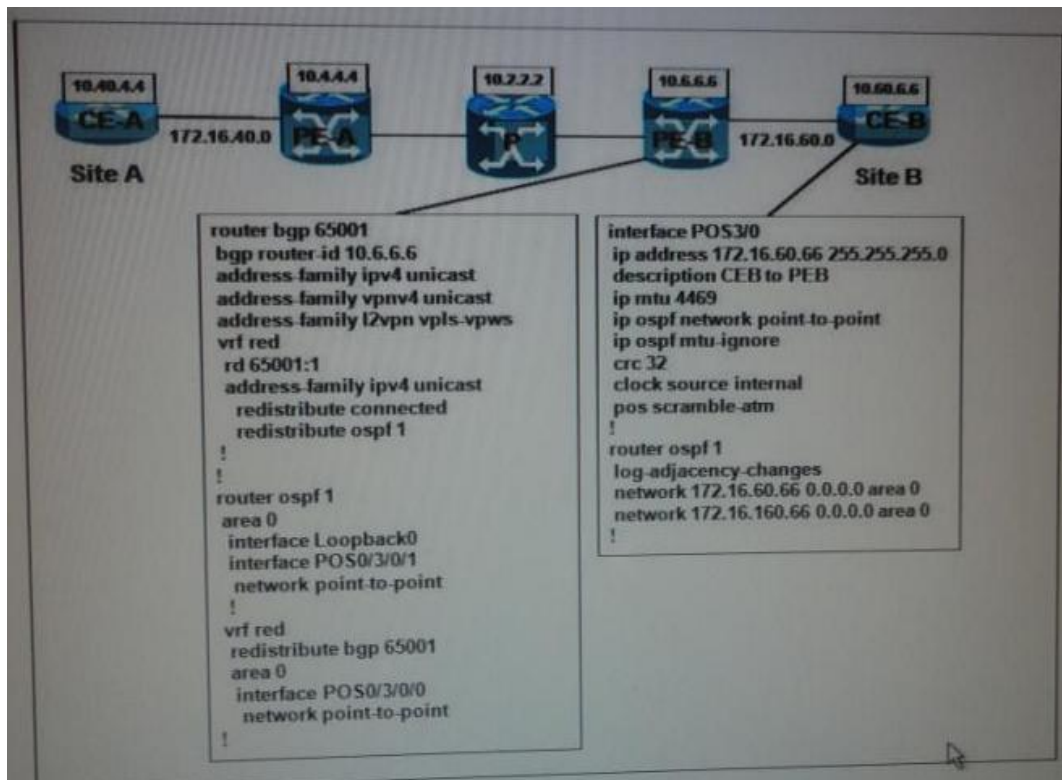
Cisco 642-780



**Maintaining Cisco Service Provider VPNs and MPLS
Networks (MSPVM)**

Version: 4.3

QUESTION NO: 1



Refer to the exhibit. The customer has an MPLS Layer 3 VPN service CE-A is not able to ping the CE-B loopback address 10.60.6.6 CE-A is able to ping the CE-B network interface address 172.16.60.66. What must be added to the configuration to allow the loopback ping to work?

- A. The PE-B configuration needs a neighbor 10.60.6.6 command added
- B. The PE-B configuration needs interface loopback0 added under vrf red area 0
- C. The CE-B configuration needs to static route added for the PE-B connected interface
- D. The CE-B configuration needs network 10.60.6.6 under router ospf 1

Answer: D

Explanation:

QUESTION NO: 2

```
ip vrf VPNA
rd 23456:100
route-target both 23456:100
route-target export 23456:300
route-target import 23456:301

ip vrf VPNB
rd 23456:200
route-target both 23456:200
route-target export 23456:300
route-target import 23456:301

ip vrf VPNC
rd 23456:300
route-target both 23456:300
route-target export 23456:301
```

Refer to the exhibit. What type of MPLS LAYER 3 VPN configurations is represented?

- A. Simple two-VPN scenario
- B. Overlapping VPNs
- C. Central services VPNs
- D. Extranet VPNs

Answer: C

Explanation:

QUESTION NO: 3

In a service provider layer 3 MPLS VPN implementations, what is the minimum number of routing on the PE routers?

- A. Three
- B. Four
- C. Five
- D. Six

Answer: A

Explanation:

QUESTION NO: 4

```
RP/0/RSP1/CPU0:VKG-3#ping 10.11.11.11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.11.11.11, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/2 ms

RP/0/RSP1/CPU0:VKG-3#ping mpls ipv4 10.11.11.11/32
Sending 5, 100-byte MPLS Echos to 10.11.11.11/32,
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/5/7 ms

RP/0/RSP1/CPU0:VKG-3#ping mpls pseudowire 10.11.11.11 100
Sending 5, 100-byte MPLS Echos to 10.11.11.11 vc: 100,
QQQQQ
Success rate is 0 percent (0/5)
```

Refer to the exhibit. The commands in the figure were executed from a cisco ASR 9000 series router. The remote end of the ping is a cisco XR12000 series router. Which statement is true?

- A. The XR 12000 has an access list that blacks the MPLS pseudowire ping
- B. The interface connected to the XR 12000 is not running LDP
- C. These results will occur during convergence when MPLS LDP sync is enabled
- D. The MPLS pseudowire ping was not sent

Answer: D

Explanation:

QUESTION NO: 5


```

debug mpls packet
ASR 9000
XR12000
CRS-1
RP/0/0/CPU0:XR12000# debug mpls packet
LC/0/0/CPU0:May 27 10:44:45.011 : netio[212]: mpls_switch-2: received an mpls packet on GigabitEthernet0_0_1_1:
proto=0, direction = ingress
LC/0/0/CPU0:May 27 10:44:45.011 : netio[212]: Flag set to 0x00000001
LC/0/0/CPU0:May 27 10:44:45.012 : netio[212]: mpls_get_payload_start: ipv4: src=0xa646402, dst=0x7f000001
LC/0/0/CPU0:May 27 10:44:45.013 : netio[212]: mpls_get_payload_start: src. port = 0xdaf, dst. port = 0xdaf, protocol =
0x11
LC/0/0/CPU0:May 27 10:44:45.013 : netio[212]: mpls_switch: GigabitEthernet0_0_1_1, mpls eos 1, ttl 1, len 132, inlabel
16017, tbl_id=0xe0000000, vrf_id=0x60000000
LC/0/0/CPU0:May 27 10:44:45.014 : netio[212]: [mpls/netio/src/mpls_netio_switch.c:1133] l3_len:[0x00000078]
readoff:[0x0000000C]
LC/0/0/CPU0:May 27 10:44:45.014 : netio[212]: Flag set to 0x00000001
LC/0/0/CPU0:May 27 10:44:45.015 : netio[212]: CW 0x46000078
LC/0/0/CPU0:May 27 10:44:45.015 : netio[212]: CW NOT found
LC/0/0/CPU0:May 27 10:44:45.016 : netio[212]: iphdr_len [24]
LC/0/0/CPU0:May 27 10:44:45.016 : netio[212]: mpls_netio_is_lspv_pkt: Packet is a LSP ping packet. UDP dest
port=3503
LC/0/0/CPU0:May 27 10:44:45.017 : netio[212]: [mpls/netio/src/mpls_netio_switch.c:1307] network_start pointing at:
0x00000046
LC/0/0/CPU0:May 27 10:44:45.017 : netio[212]: First 4 network bytes: 0x46000078
LC/0/0/CPU0:May 27 10:44:45.017 : netio[212]: mpls_switch: MPLS TTL expired on LSP ping packet. Send to ipv4
netio for processing
LC/0/0/CPU0:May 27 10:44:45.030 : netio[212]: mpls_switch-2: received an mpls packet on GigabitEthernet0_0_1_1:
proto=0, direction = ingress
LC/0/0/CPU0:May 27 10:44:45.030 : netio[212]: Flag set to 0x00000001
LC/0/0/CPU0:May 27 10:44:45.031 : netio[212]: mpls_get_payload_start: ipv4: src=0xa646402, dst=0x7f000001
LC/0/0/CPU0:May 27 10:44:45.031 : netio[212]: mpls_get_payload_start: src. port = 0xdaf, dst. port = 0xdaf, protocol =
0x11
LC/0/0/CPU0:May 27 10:44:45.032 : netio[212]: mpls_switch: GigabitEthernet0_0_1_1, mpls eos 1, ttl 2, len 132, inlabel
16017, tbl_id=0xe0000000, vrf_id=0x60000000
LC/0/0/CPU0:May 27 10:44:45.032 : netio[212]: mpls_switch: DISPOSITION (PHP) to ip: prop. ttl = 1, ttl=1

```

Refer to the exhibit. The output from the debug command is from the Cisco XR12000 series Router, which statement is true?

- A. The output is the result of single iteration of mpls ping executed on the cisco ASR 9000 Series Router'
- B. The output is the result of single iteration of mpls ping executed on the cisco ASR 9000 series Router
- C. The packets did not reach its destination because the control word was not found
- D. The output is the result of single iteration of mpls ping executed on the cisco CRS-1 carrier routing system.

Answer: C

Explanation:

QUESTION NO: 6

```

RP/0/0/CPU0:GSR-1#show cef vrf red 10.22.22.22 detail
10.22.22.22/32, version 1, internal 0x40040001 (0x9cdb88f8) [1], 0x0 (0x0), 0x4100 (0x9d3658f0)
Updated May 26 14:01:44.060
Prefix Len 32, traffic index 0, precedence routine (0)
gateway array (0x9cc5b668) reference count 5, flags 0x80700, source rib (3),
[1 type 1 flags 0x901101 (0x9d6092e4) ext 0x0 (0x0)]
LW-LDI[type=0, refc=0, ptr=0x0, sh-ldi=0x0]
via 192.168.252.3, 5 dependencies, recursive
next hop 192.168.252.3 via 16000/0/21
next hop 10.100.100.2   Gi0/0/1/1   labels imposed [ImplNull 143989]

Load distribution: 0 (refcount 1)

Hash OK Interface Address
0 Y Unknown 16000/0

RP/0/0/CPU0:GSR-1#show bgp vpnv4 unicast labels
BGP router identifier 192.168.253.1, local AS number 65000
<snip>
Status codes: s suppressed, d damped, h history, * valid, > best
i - internal, S stale
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Rcvd Label Local Label
Route Distinguisher: 65000:9 (default for vrf red)
>i10.1.1.1/32 192.168.252.3 143988 no label
>i10.9.9.0/31 192.168.252.3 143989 no label
> 10.109.109.109/32 0.0.0.0 no label 16014
>i10.170.170.0/24 192.168.252.3 143990 no label
>i10.22.22.22/32 192.168.252.3 143989 no label
>i192.1.99.0/24 192.168.252.3 143991 no label
> 192.168.99.0/24 0.0.0.0 no label 16014

```

Refer to the exhibit. The commands were executed on a Cisco XR12000 PE. Which statement is true?

- A. The outer label for prefix 10.22.22.22 is 143989.
- B. The outer label for prefix 10.22.22.22 is 16000.
- C. The outer label for prefix 10.22.22.22 is implNull.
- D. Prefix 10.22.22.22 has no outer label.

Answer: D

Explanation:

QUESTION NO: 7

You have configured a MPLS-TE tunnel with a predefined explicit path as primary and dynamic path as the backup. The tunnel was designed to carry customer traffic from site A to site B. although the MPLS-TE tunnel is set up. It is not carrying the traffic. Traffic has to go to network 171.68.0.0/16 connected to site B.

Which three show commands would you use to identify and resolve the issue?

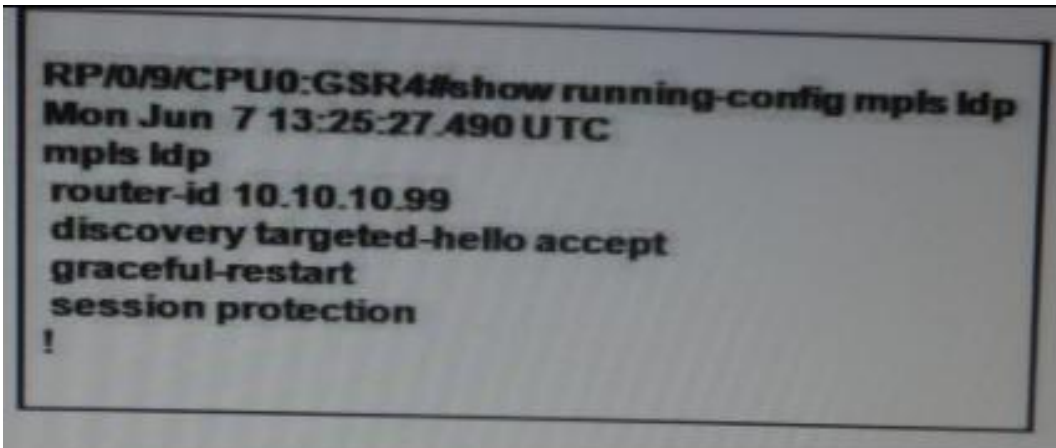
- A. Show mpls traffic-eng-link-management bandwidth to check the bandwidth is available.
- B. Show ip cef network-number to check if tags are imposed
- C. show mpls traffic-eng topology igp-id ospf NETWORK-NUMBER brief to check routing issue
- D. Show mpls traffic-eng autoroute to check auto routing is enabled
- E. Show mpls forwarding-table LABEL-number details to outgoing tags

F. Show ip cef tunnel1 to check traffic is passing through the tunnel

Answer: B,D,F

Explanation:

QUESTION NO: 8



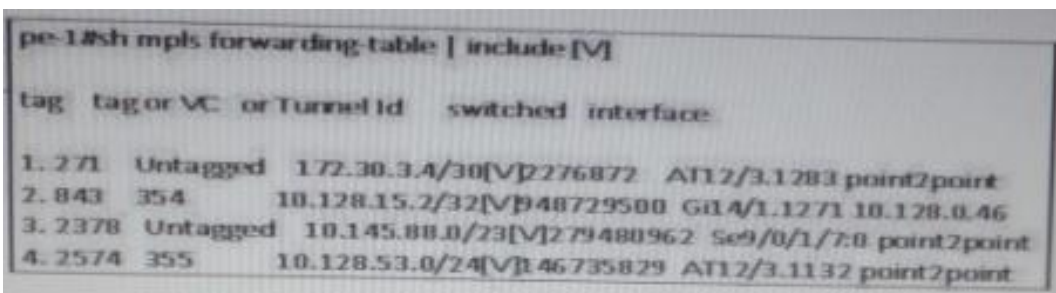
Refer to the exhibit. Is nonstop forwarding enabled or disabled, and which part of show command can be used to determine this?

- A. Enabled, graceful-restart
- B. Enabled, session protection
- C. Disabled, graceful-restart
- D. Disabled, session protection

Answer: A

Explanation:

QUESTION NO: 9



Refer to the exhibit. What two problems are associated with this show MPLS forwarding command

output; given that this command only shows forwarding information for CE's faces interfaces and associated statistics? (Choose two)

- A. Lin 1 and 3 indicate the problem
- B. Lin 2 and 4 indicate the problem
- C. LDP is not active on AT 12/3.1283 and se 9/0/1/7:0 that is why packets are sent as IP packets and no MPLS packets
- D. There is a possible routing loop which makes pe-1 a transit router for the VRFs which are assigned to Gi14/1.1271 and AT12/3.1132
- E. There is label corruption for VRFs where Gi14/1.1271 and AT12/3.1132 belong to, the value of a locally assigned label cannot be more than the value of outgoing label

Answer: B,E

Explanation:

QUESTION NO: 10

You have been assigned the task of collecting the relevant information for trouble shooting a Layer 3 MPLS VPN routing issue across two (same VPN) for a customer. After initial diagnostics you have ruled out any connectivity issues and any hardware or software bugs.

Which four steps should you use to collect information for troubleshooting the VPN routing issue? (Choose four)

- A. Redistribution of routes into MP-BGP
- B. Route filters that might prevent routes from propagating
- C. Interface state between PE and CE devices
- D. VRF route target import and export details
- E. MPLS forwarding state between PE and core routers
- F. VRF route limit on the PE devices
- G. Interface error counters on the customer-facing PE interface

Answer: A,B,D,F

Explanation:

QUESTION NO: 11

Which three MPLS services require MPLS layer 2 VPNs? (Choose three)

- A. Any-to-any interworking

- B. Central services VPN
- C. VPLS
- D. VPWS

Answer: A,C,D

Explanation:

QUESTION NO: 12

A service provider is running OSPF and has turned the network for fast convergence. The service provider has determined that its routers should also be configured for LDP synchronization. If the service provider follows ITIL-based practices, how should this change be treated?

- A. As a standard change
- B. As a minor change
- C. As a major change
- D. As an emergency change

Answer: C

Explanation:

QUESTION NO: 13

What are three basic functional requirements when configuring MPLS layer 2 VPLS?

(Choose three)

- A. A CE to PE routing protocol must be used to exchange routing prefixes
- B. A full mesh of pseudowire (emulated VC) connections must be used between PE routers
- C. Each PE must have a VFI configured
- D. LDP protocol must be running
- E. Auto discovery is needed to identify endpoints outside of the VPLS domain.

Answer: B,C,D

Explanation:

QUESTION NO: 14

```

PE#show ip bgp vpnv4 all

BGP table version is 17, local router ID is 192.168.1.17
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal, r RIB-failure, S Stale

Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop        Metric LocPrf Weight Path
Route Distinguisher: 1:10 (default for vrf Customer_1)
*> 10.1.11.16/28     150.1.11.17          10           32768 ?
*> 10.1.11.49/32     150.1.11.17          10           32768 ?
*> 10.1.11.64/28     150.1.11.17          10           32768 ?
*>i10.1.12.16/28     192.168.1.33         1    100        0 ?
*>i10.1.12.49/32     192.168.1.33         1    100        0 ?
*>i10.1.12.64/28     192.168.1.33         1    100        0 ?
*> 150.1.11.16/28    0.0.0.0              0           32768 ?
*>i150.1.12.16/28    192.168.1.33         0    100        0 ?

PE#show ip route vrf Customer_1

Routing Table: Customer_1
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
B    10.1.11.16/28 [90/2195456] via 150.1.11.17, 01:32:43, Serial0/0.101
B    10.1.12.16/28 [200/1] via 192.168.1.33, 01:31:16
B    10.1.11.49/32 [90/2297856] via 150.1.11.17, 01:32:43, Serial0/0.101
B    10.1.12.49/32 [200/1] via 192.168.1.33, 01:31:16
B    10.1.11.64/28 [90/2297856] via 150.1.11.17, 01:32:43, Serial0/0.101
B    10.1.12.64/28 [200/1] via 192.168.1.33, 01:31:16
B    150.1.0.0/28 is subnetted, 2 subnets
B    150.1.12.16 [200/0] via 192.168.1.33, 01:31:16
C    150.1.11.16 is directly connected, Serial0/0.101

```

Refer to the exhibit. Which statement about the two show commands output is true?

- A. The CE-PE routing protocol between the PE and the Customer_1 CE router is BGP.
- B. The PE router has learned four prefixes over the multiprotocol BGP session.
- C. 192.168.1.33 is the CE router and 150.1.11.17 is the PE router.
- D. The PE router is supporting more than one MPLS Layer 3 VPN customers.

Answer: B

Explanation:

QUESTION NO: 15

What are four essential parts of the MP-BGP VPNv4 update? (Choose four)

- A. LDP label
- B. VPN label
- C. IPv4 prefix with route descriptor
- D. LSP identifier
- E. BGP next-hop address
- F. Route targets (extended communities)

Answer: B,C,E,F

Explanation:

QUESTION NO: 16

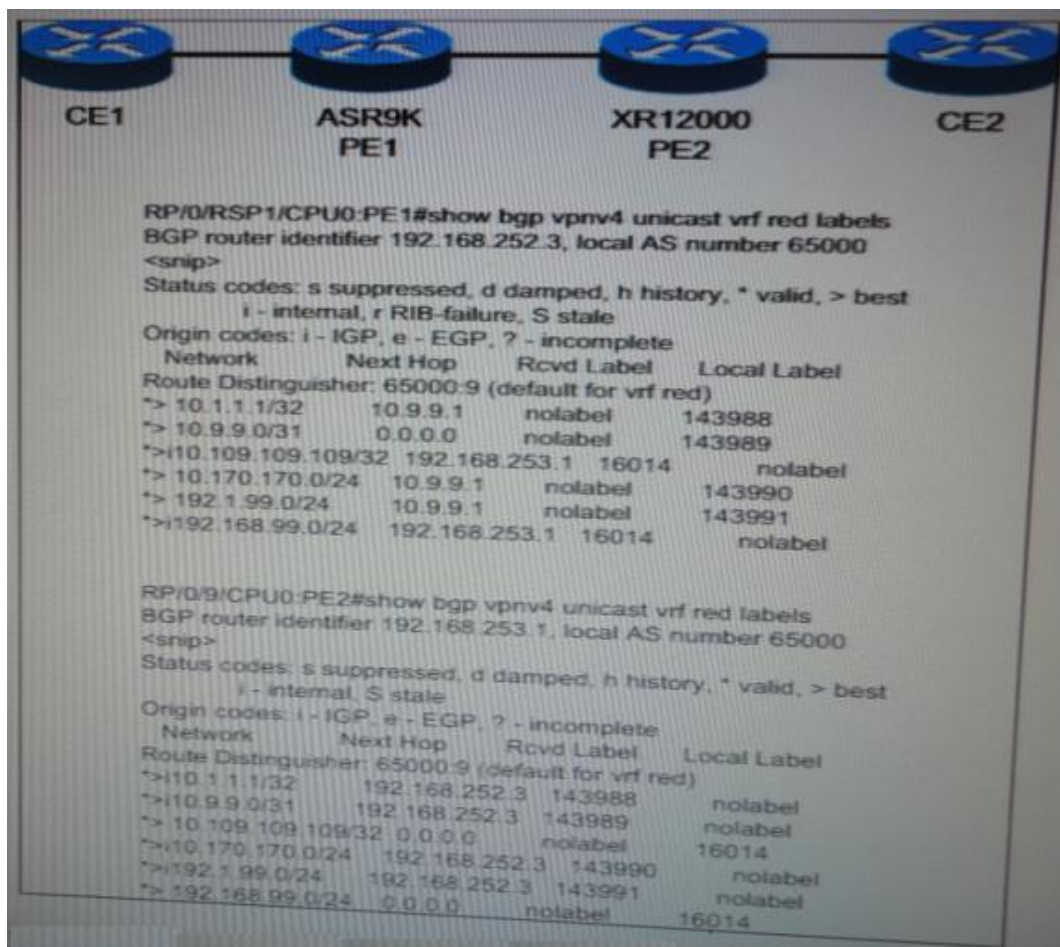
What function does MP-BGP provides that is required for MPLS layer 3 VPM services?

- A. MP-BGP modifies or swaps the LDP label to indicate the next-hop router in MPLS cloud
- B. MP-BGP provides the penultimate hop popping function
- C. MP-BGP performs the standard MPLS LDP functions in layer 3 environment
- D. MP-BGP advertises the VPN label and identifies the correct egress PE interface

Answer: D

Explanation:

QUESTION NO: 17



Refer to the exhibit. A layer 3 VPN is configured between PE1 and PE2. CE1 is connected to PE1. CE2 is connected to PE2. CE1 and CE2 are in VRF red. From which router does the prefix 10.170.170.0/24 originates?

- A. CE1
- B. PE1
- C. PE2
- D. CE2

Answer: A

Explanation:

QUESTION NO: 18

What two mechanisms are used to improve both MPLS network performance and availability?
(Choose two)

- A. LDP and BGP synchronization
- B. Stateful switchover with fast Reroute
- C. Stateful switchover
- D. BGP timer tuning
- E. Stateful switchover with NSR

Answer: D,E

Explanation:

QUESTION NO: 19

A cisco 7600 series Router is experiencing high CPU utilization and poor forwarding performance. Investigation reveals that packet forwarding has been punted to the RP. What are two possible causes of this behavior? (Choose two)

- A. The routing table is too large
- B. Liberal retention mode has been disabled
- C. An insufficient number of flow masks are configured
- D. A large number of access list entries are configured
- E. IGP timers need to be returned to improve routing convergence

Answer: A,D

Explanation:

QUESTION NO: 20

Layer 3 VPN is the primary service running on your network. You have POP locations and recently

you have seen memory and CPU utilization go above 90% in two out of four PE devices in one POP location. You have maximum-route-limit configured for each VPN.

What additional parameter should be monitored proactively to address these issues?

- A. Number of prefixes per VRF route in the router
- B. Number of VRFs configured on each Router
- C. Number of route targets for each VRF
- D. LDP neighbor statistics for any flaps
- E. Packets in and out for each VRF

Answer: B

Explanation:

QUESTION NO: 21

Which MPLS layer 2 VPN services strip 802.1q tags before encapsulating the frame?

(Choose two)

- A. Ethernet over MPLS port mode
- B. Ethernet over MPLS VLAN mode
- C. MPLS Bridging
- D. MPLS VPLS

Answer: B,D

Explanation:

QUESTION NO: 22

```
PE11#show mpls ldp bindings
tib entry: 150.1.11.16/28, rev 6
  local binding: tag: imp-null
  remote binding: tsr: 192.168.1.81:0, tag: 17
tib entry: 150.1.11.32/28, rev 8
  local binding: tag: imp-null
  remote binding: tsr: 192.168.1.81:0, tag: 18
tib entry: 150.1.12.16/28, rev 20
  local binding: tag: 21
  remote binding: tsr: 192.168.1.81:0, tag: 22
tib entry: 150.1.12.32/28, rev 22
  local binding: tag: 22
  remote binding: tsr: 192.168.1.81:0, tag: 23
tib entry: 192.168.1.17/32, rev 4
  local binding: tag: imp-null
  remote binding: tsr: 192.168.1.81:0, tag: 16
tib entry: 192.168.1.33/32, rev 18
  local binding: tag: 20
  remote binding: tsr: 192.168.1.81:0, tag: 21
tib entry: 192.168.1.48/28, rev 2
  local binding: tag: imp-null
  remote binding: tsr: 192.168.1.81:0, tag: imp-null
tib entry: 192.168.1.64/28, rev 16
  local binding: tag: 19
```

PE11#

Refer to the exhibit. Which two statements are true? (Choose two)

- A. 150.1.12.16 is directly connected subnet
- B. The router is advertising label 20 destinations 192.168.1.33.
- C. The upstream P router is to perform PHP for destinations on 192.168.1.48
- D. The forward data to device on the network 150.1.12.32 the router will remove label 22 and impose label 23

Answer: B,C

Explanation:

QUESTION NO: 23

```

RP/0/0/CPU0:XR12000#show bgp vprn4 unicast vrf red labels
BGP router identifier 192.168.253.1, local AS number 65000
BGP generic scan interval 60 secs
BGP table state: Active
Table ID: 0x0
BGP main routing table version 87
BGP scan interval 60 secs

Status codes: s suppressed, d damped, h history, * valid, > best
               i - internal, S stale
Origin codes: i - IGP, e - EGP, ? - incomplete
   Network        Next Hop        Rcvd Label        Local Label
Route Distinguisher: 65000:9 (default for vrf red)
*>i10.1.1.1/32      192.168.252.3      143988            no-label
*>i10.9.9.0/31      192.168.252.3      143989            no-label
*> 10.109.109.109/32 0.0.0.0            no-label          16014
*>i10.170.170.0/24  192.168.252.3      143990            no-label
*>i10.22.22.22/32   192.168.252.3      143989            no-label
*> 10.0.65.0/24     192.168.99.1        no-label          16016
*>i192.1.99.0/24    192.168.252.3      143991            no-label
*> 192.168.99.0/24  0.0.0.0            no-label          16014

```

Refer to the exhibit. The command was executed on cisco XR12000 series router. The router is directly connected to a remote PE, which has router ID of 192.168.253.3. A single CE is connected to the remote PE. Based on the command output, which label allocation mode is remote PE running?

- A. Per-prefix
- B. Per-ce
- C. Per-vrf
- D. Per-pe

Answer: A

Explanation:

QUESTION NO: 24

```

RP/0/RSP1/CPU0:VKG-3#show mpls ldp bindings 192.168.253.1/32
192.168.253.1/32, rev 537
Local binding: label: 16004
Remote bindings: (1 peers)
  Peer                Label
  -----
  192.168.253.1:0     IMP-NULL

```

Refer to the exhibit. The command in the diagram was executed from a cisco ASR 9000 Series router. Which statement is true?

- A. 192.168.253.1 is router ID of the ASR9000
- B. The ASR 9000 will use label 16004 to reach 192.168.253.1
- C. The ASR 9000 will perform penultimate hop popping on packets destined to 192.168.253.1
- D. The ASR 9000 advertises a label of IMP-NULL to its peers for 192.168.253.1 reachability

Answer: C

Explanation:

QUESTION NO: 25

You have implemented MPLS-TE on the core and enabled Fast Reroute. BFD is configured on the core facing links.

Every thing was working normally until last month when you added video service on the network. Because of business requirements, video packets are treated as high priority and 70% of bandwidth is allocated for the low latency queue.

Recently you noticed BFD session losses and MPLS-TE tunnel flaps on the network. Which parameter should you have mentioned proactively to avoid this issue?

- A. BFD session timers and hello message
- B. MPLS-TE tunnel bandwidth and utilization
- C. QoS drops for each class of service on core-facing links
- D. CPU utilization on the PE routers
- E. Latency and jitter between PE routers

Answer: C

Explanation:

QUESTION NO: 26

Which decode corresponds correctly to this MPLS header. 0x00 0x04 0xAB 0xFE?

- A. Label: 0004, Exp AB. S:1.TTL:7E
- B. Label: 0004A. TTL:BF.Exp:8. S:0
- C. Label: 0004A , Exp: 5, S: 1, TTL: FE
- D. Label: 0004. TTL:AB.Exp:7E.S:1

Answer: C

Explanation:

QUESTION NO: 27

What four layer 2 MPLS VPN parameters are collected from a customer service record database to troubleshoot a VPN connectivity issue? (Choose four)

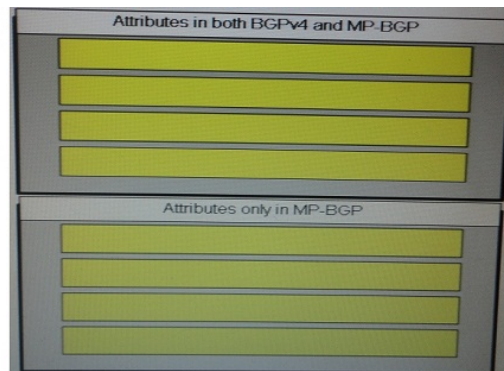
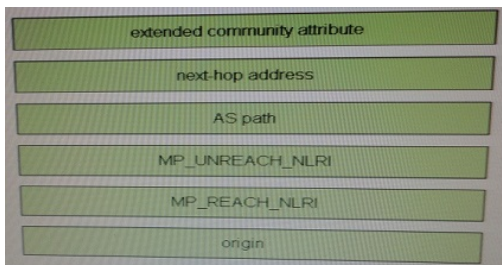
- A. Attachment circuit information
- B. UNI interface encapsulation
- C. Hop-by-hop LDP information
- D. Virtual circuit ID information
- E. Unique service identifier
- F. PE loopback information
- G. Layer 2 MPLS VPN service type

Answer: A,B,D,G

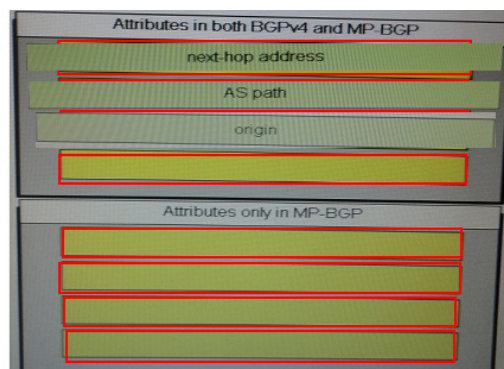
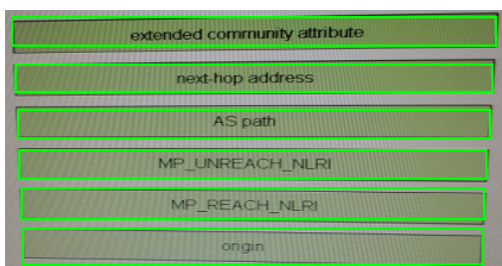
Explanation:

QUESTION NO: 28 DRAG DROP

Drag the attributes on the left to the appropriate categories on the right. Note that not all slots may be uses.



Answer:



Explanation: extended community attribute Attributes only in MP-BGP
next-hop address Attributes both in BGPv4 and MP-BGP
AS path Attributes both in BGPv4 and MP-BGP
MP_UNREACH_NLRI Attributes only in MP-BGP
MP_REACH_NLRI Attributes only in MP-BGP
Origin Attributes both in BGPv4 and MP-BGP

QUESTION NO: 29

The SP NOC has determined that a significant number of broadcast packets are being received from customer A. which could be a DoS attack.

Customer A is using MPLS layer VPLS services the exercise broadcast traffic is causing degrading performance and over utilizing the access links. Which of the cisco IOS XR MPLS layer 2 configuration parameters can be used to mitigate the problem?

- A. The storm control parameter under the 12vpn bridge group
- B. The storm control parameter under the 12vpn xconnect group
- C. The limit-broadcast parameter under the 12vpn bridge group
- D. The storm control parameter under the 12vpn xconnect

Answer: A

Explanation:

QUESTION NO: 30

A customer has stated that they can not access network 10.450.16.0/24 from one of there sites (Site C). The customer has a layer 3 MPLS VPN between three sites (site A, site B, site C) connected in full mesh topology. They do not have any issues with accessing other network addresses located in site A and site B.

Assuring the PE devices are running Cisco IOS, what are the two troubleshooting commands you should execute on the PE router connected to site C to isolate the issue? (Choose two.)

- A. Show ip vrf VRFNAME detail to check the route targets
- B. Show ip vrf VRFNAME detail to check the vrf routes
- C. show mpls l2transport binding to check local and remote labels
- D. Show MPLS interfaces INTERFACENAME detail to check MTU size
- E. Show ip bgp vpnv4 vrf VRFNAME to check advertised VPNv4 prefixes

Answer: B,E

Explanation:

QUESTION NO: 31

```

P/0/9/CPU0:XR12000#show mpls traffic-eng tunnels
Signalling Summary:
  LSP Tunnels Process: running
  RSVP Process: running
  Forwarding: enabled
  Periodic reoptimization: every 3600 seconds, next in 722 seconds
  Periodic FRR Promotion: every 300 seconds, next in 197 seconds
  Auto-bw enabled tunnels: 0 (disabled)

SP Tunnel 192.168.252.3 1 [2] is signalled, connection is up
Tunnel Name: VKG-3_tl Tunnel Role: Mid
InLabel: GigabitEthernet0/0/1/1, 16003
OutLabel: TenGigE0/1/1/0, implicit-null
Signalling Info:
  Src 192.168.252.3 Dst 192.168.254.1, Tun ID 1, Tun Inst 2, Ext ID 192.168.252.3
  Router-IDs: upstream 192.168.252.3
               local 192.168.253.1
               downstream 192.168.254.1
Path Info:
  Incoming Address: 10.100.100.1
  Incoming Explicit Route:
    Strict, 10.100.100.1
    Strict, 10.200.200.0
    Strict, 192.168.254.1
  Outgoing:
  Explicit Route:
    Strict, 10.200.200.0
    Strict, 192.168.254.1
  Record Route: None
  Tspec: avg rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
  Session Attributes: Local Prot: Not Set, Node Prot: Not Set, BW Prot: Not Set
Resv Info:
  Record Route: None
  Fspec: avg rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
Displayed 0 (of 0) heads, 1 (of 1) midpoints, 0 (of 0) tails
Displayed 0 up, 0 down, 0 recovering, 0 recovered heads

```

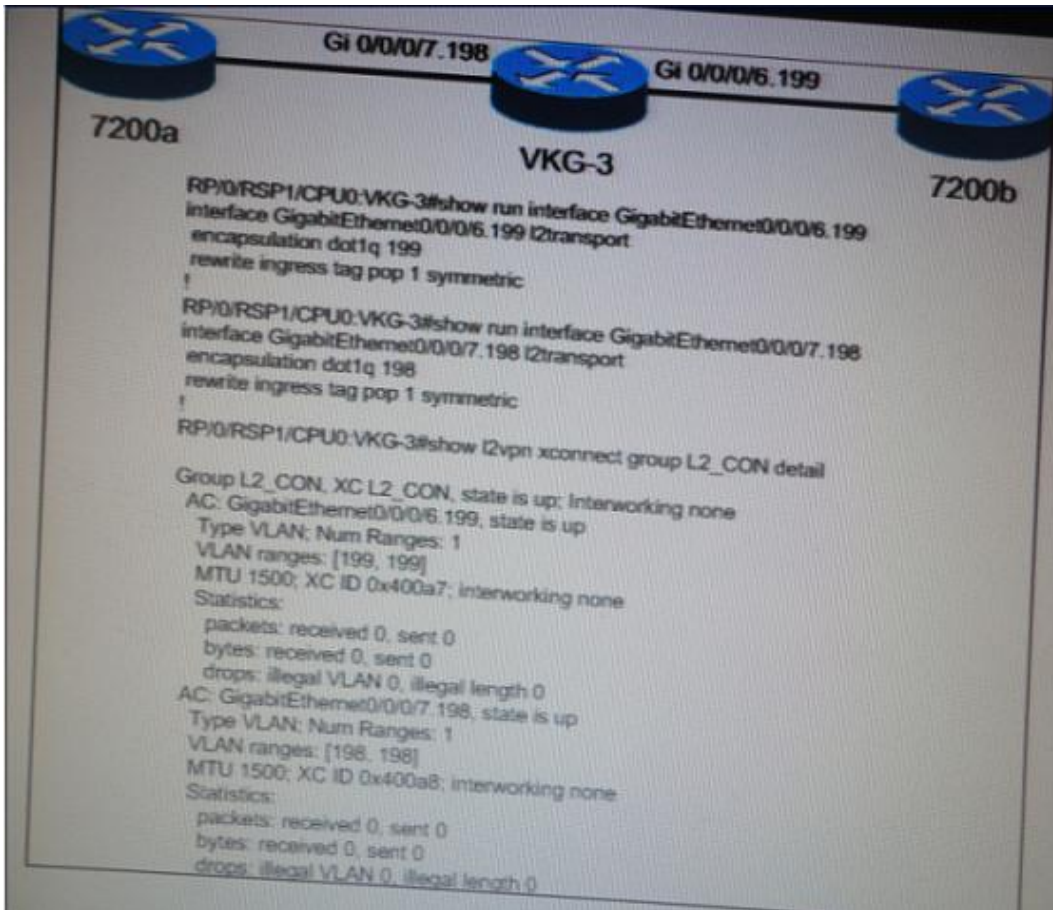
Refer to the exhibit. The command was executed on a cisco XR12000 series router. Which statement about the tunnel displayed is true?

- A. The command was executed on the head end of the tunnel
- B. A label of 16003 will be used to reach the next hop for the tunnel
- C. 10.200.200.0 is the IP address of remote interface of the next downstream hop for the tunnel.
- D. The command was executed on the tail end of the tunnel

Answer: D

Explanation:

QUESTION NO: 32



Refer to the exhibit. The commands were executing from VKG-3, which is the Cisco ASR 9000 Series router. Which statement is true?

- A. Layer 2 reachability can be achieved between 7200a and 7200b with the current configuration on VKG-3
- B. Layer 2 reachability between 7200a and 7200b will fail because of VLAN mismatch configured on VKG-3
- C. Layer 2 reachability between 7200a and 7200b will fail because VKG-3 needs "rewrite egress tag push 1 symmetric" configured under both interfaces
- D. Layer 2 reachability between 7200a and 7200b will fail because the MTU needs to be increased 1504 to include for the VLAN header file

Answer: A

Explanation:

QUESTION NO: 33

What are three characteristics that define MP-BGP performance? (Choose three)

- A. Convergence upon failure and recovery

- B. Time to establish a tunnel
- C. Import and export of VPNv4 routes
- D. Time to complete a SPF calculation
- E. Route table scanning for changes
- F. Time to synchronize with LDP

Answer: A,C,E

Explanation:

QUESTION NO: 34

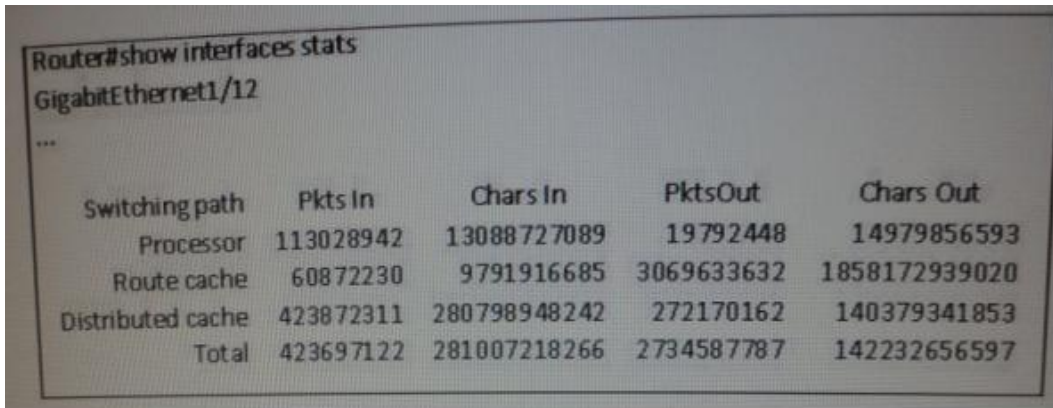
```
vrf VPNA
address-family ipv4 unicast
maximum-prefix 1000 75
!
!
router eigrp 23456
vrf VPNA
address-family ipv4
maximum-prefix 1000 75
!
!
router bgp 23456
vrf VPNA
neighbor 10.1.30.1
address-family ipv4 unicast
maximum-prefix 1000 75
!
!
!
```

Refer to the exhibit. Which statement best describes the Cisco IOS XR VRF protection that is configured?

- A. The maximum number of configured VPNs is limited to 1000 with an active limit of 75
- B. The number of prefixes for VPNs number 1000 is limited to 75
- C. The number of VRF table entries is limited to 1000, and a syslog message will be sent at 75%
- D. The maximum number of pseudowire VCs is limited to 1000 and a syslog message will be sent at 75%

Answer: C

Explanation:

QUESTION NO: 35

```
Router#show interfaces stats
GigabitEthernet1/12
***
Switching path  Pkts In    Chars In    PktsOut     Chars Out
Processor       113028942   13088727089 19792448    14979856593
Route cache     60872230    9791916685  3069633632  1858172939020
Distributed cache 423872311   280798948242 272170162   140379341853
Total           423697122   281007218266 2734587787  142232656597
```

Refer to the exhibit. The show interface statistic was taken five minutes after the interface counter was cleared. What problem is indicated in the exhibit?

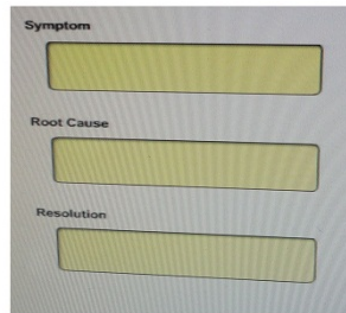
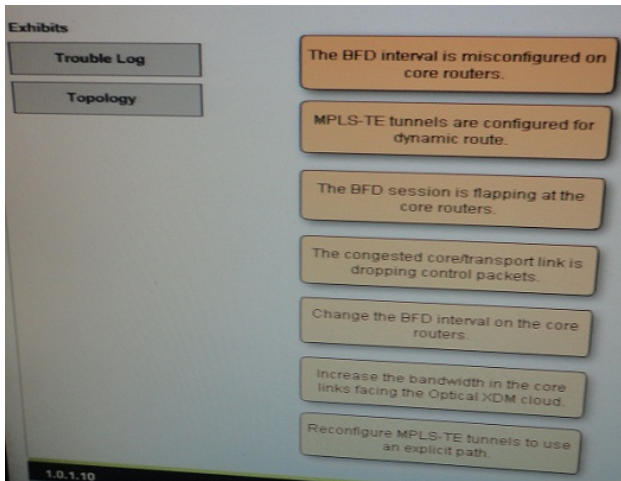
- A. The hardware forwarding capacity of the router has been exceeded
- B. The IGP and LDP of router need to be synchronized
- C. The RP has just failed over from the primary to standby
- D. The IGP timers of the router need to be turned

Answer: A

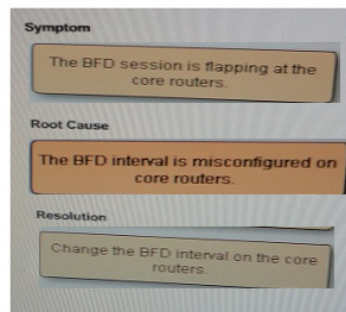
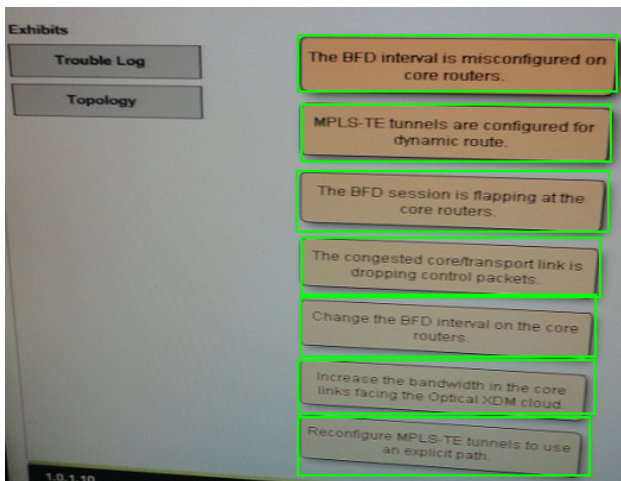
Explanation:

QUESTION NO: 36 DRAG DROP

Refer to Exhibit. MPLS-TE is configured in the core and BFD is used for faster link failure detection. Currently the MPLS-TE tunnels are flapping. Identify the Symptom of the flapping, the root cause of the issue, and the resolution. Click and drag the elements on the left to the correct identifier on the right. It is possible that you will not use all the elements.



Answer:



Explanation:

- The BFD Session is flapping at core routers = Symptom
- The BFD interval is misconfigured on core routers = Root Cause
- Change the BFD interval on core routers = Resolution

QUESTION NO: 37

```
RP/0/RSP1/CP00:VKG-3#show l2vpn bridge-domain bdl-name GROUP5 detail
<- snip ->
List of Access PWs:
List of NIs:
VNI group5
PW neighbor 192.168.253.1, PW ID 100, state is down ( remote standby )
PW class not set, XC ID 0x1ffc0004
PW encapsulation MPL3, protocol LDP
PW type Ethernet, control word disabled, interworking none
PW backup disable delay 0 sec
sequencing not set

MPL3:
-----
Local                               Remote
Label                               16006
Group ID                             0x10
Interface                             group5
MTU                                   1500
Control word                         disabled
PW type                             Ethernet
MPL3 CV type                         0x2
MPL3 CC type                         0x0
                                   (LSP ping verification)
                                   (router alert label)
                                   (LSP copy)

<- snip ->
RP/0/RSP1/CP00:VKG-3#show mpls forwarding
Local Outgoing Prefix Outgoing Next Hop Bytes
Label Label or ID Interface Interface Hop Switched
-----
16000 Pop PW(192.168.253.1:801) Gi0/0/0/7.801 point2point 136941744
16001 Pop PW(192.168.253.1:100) Gi0/0/0/7.100 point2point 0
16002 Pop PW(192.168.253.1:501) Gi0/0/0/7.501 point2point 612700210
16003 Pop PW(192.168.253.1:1501) Gi0/0/0/7.1501 point2point 0
16004 Unlabelled 192.168.253.1/32 Gi0/0/0/7.1501 point2point 0
16006 Pop PW(192.168.253.1:1701) Gi0/0/0/7.1701 point2point 0
16008 Pop PW(192.168.253.1:1401) Gi0/0/0/7.1401 point2point 4629306
```

Refer to the exhibit. What could cause the pseudowire for bridge group GROUP5 to be down?

- A. An interface along the LSP is not running LDP
- B. The PW type needs to be changed to be VLAN mode
- C. The MTU needs to be increased by four to include the VLAN header
- D. The local and remote labels do not match

Answer: A

Explanation:

QUESTION NO: 38

Which two statements describe when deployment of IP SLA could be regarded as standard or minor change as defined by ITIL? (Choose two)

- A. A low-frequency probe is being placed on one router for limited time.
- B. A high-frequency probe is being placed on one router for limited time.
- C. A low-frequency probe is being deployed to a few routers for a limited time.
- D. A low-frequency probe is being deployed to a few routers permanently
- E. A low-frequency probe is being deployed across the network for limited time
- F. A low-frequency probe is being deployed across the network permanently

Answer: A,C

Explanation:

QUESTION NO: 39

```

pel#trace mpls ipv4 14.14.14.14 255.255.255.255
Tracing MPLS Label Switched Path to 14.14.14.14/32, timeout is 2 seconds

Codes: '.' - success, 'Q' - request not sent, '.' - timeout,
'L' - labeled output interface, 'B' - unlabeled output interface,
'D' - DS Map mismatch, 'F' - no FEC mapping, 'f' - FEC mismatch,
'M' - malformed request, 'm' - unsupported tlvs, 'N' - no label entry,
'P' - no rx intf label prot, 'p' - premature termination of LSP,
'R' - transit router, 'I' - unknown upstream index,
'X' - unknown return code, 'x' - return code 0

Type escape sequence to abort.
 0 37.37.37.2 MRU 4072 [Labels: 66 Exp: 0]
 I 1 37.37.37.1 MRU 4072 [Labels: 99 Exp: 0] 96 ms
 I 2 78.78.78.2 MRU 9102 [Labels: 99 Exp: 0] 76 ms
 I 3 181.181.181.1 MRU 4072 [Labels: 66 Exp: 0] 116 ms
 I 4 110.110.110.1 MRU 1500 [Labels: 66 Exp: 0] 88 ms
 I 5 103.103.103.2 MRU 4076 [Labels: implicit-null Exp: 0] 68 ms
 I 6 135.135.135.2 80 ms
  
```

Refer to the exhibit. Some of your customers are reporting poor network performance when they use FTP or SFTP to transfer large data files across your MPLS VPN core. Recently, some backbone links in your MPLS VPN core have been upgraded from POS OC-48 to the new 10 Gigabit Ethernet. You have collected the output of the traceroute mpls ipv4 command with the remote PE as a destination.

All devices are running on Cisco IOS. What must be fixed in the network to resolve the issue?

- A. MPLS should be enabled on router ID 103.103.103.2.
- B. MTU on 78.78.78.2 has to be reduced to 4072.
- C. Exp is set to 0, which causes FTP traffic to be dropped.
- D. MTU on 110.110.110.1 should be changed to 4072.
- E. No route exists to 14.14.14.14 on 135.135.135.2.

Answer: D

Explanation:

QUESTION NO: 40

You have designed and implemented MPLS-TE for fast router in your network core. Your network

core consists of Cisco CRS-1 carrier routing System devices interconnected using XDM and Sonet Transport. You are using OSPF as the IGP, and MP-BGP is running on the PE devices to support Layer 3 VPN services. You are offering voice service over VPN circuits. So you want a link convergence of no greater than 100ms.

Which option should you use for faster link failure detection and to improve the performance of your network?

- A. Use OSPF fast hello packets with dead-interval and hello multiplier
- B. Reduce MP-BGP hello timers between the neighbors
- C. Configure MPLS-TE tunnels with the dynamic backup option
- D. Configure MPLS-TE tunnels to autoroute announce
- E. Configure BFD on the core links with the appropriate BFD multiplier.

Answer: E

Explanation:

QUESTION NO: 41

Which two IP SLA probe types can be used to monitor MPLS and VPNS for SLA compliance? (Choose two)

- A. UDP echo
- B. UDP echo ping
- C. MPLS LSP ping
- D. UDP delay
- E. VRF
- F. HTTP

Answer: C,E

Explanation:

QUESTION NO: 42

```
interface loopback0
ip address 10.10.10.10 255.255.255.255
!
router bgp 65001
 neighbor 209.165.200.230 remote-as 65001
 neighbor 209.165.200.230 update-source loopback0
!
 address-family vpnv4
  neighbor 209.165.200.230 activate
  neighbor 209.165.200.230 next-hop-self
  neighbor 209.165.200.230 send-community both
```

Refer to the exhibit. Given the partial configuration, which statement is true?

- A. The 209.165.200.230 router is a P router
- B. The 209.165.200.230 router is a CE router
- C. The 209.165.200.230 router is a PE router
- D. The 209.165.200.230 router is a customer oriented behind the CE router

Answer: C

Explanation:

QUESTION NO: 43

What happens to a captured ingress MPLS packet that has the header 0x00 0x00 0x7B 0x01?

- A. The packet will be dropped because it uses an incorrect label
- B. The packet will be dropped because it reached its TTL limit
- C. The packet will be dropped because its EXP value is corrupted
- D. The packet will be MPLS switched because nothing is wrong with it

Answer: B

Explanation:

QUESTION NO: 44 DRAG DROP

Drag the list of commands on the left to the proper command categories on the right.

	Global Commands Only
mpls ip	
mpls traffic engineering tunnels	
ip cef	
ip rsvp bandwidth	
ip vrf forwarding	

	Interface Command Only

	Both Global and Interface Commands

Answer:

	Global Commands Only
mpls ip	ip cef
mpls traffic engineering tunnels	
ip cef	
ip rsvp bandwidth	
ip vrf forwarding	

	Interface Command Only
	mpls ip
	ip rsvp bandwidth
	ip vrf forwarding

	Both Global and Interface Commands
	mpls traffic engineering tunnels

Explanation:

mpls ip Interface Commands Only

mpls traffic engineering tunnels Both Global & Interface commands

ip cef Global Commands Only

ip rsvp bandwidth Interface Commands Only

ip vrf forwarding Interface Commands Only

QUESTION NO: 45

Which four tasks are used to configure MPLS? (Choose four)

- A. Create a VRF instance for the VPN
- B. Assign customer interfaces (access links) to a VPN
- C. Enable the VFI and MPLS encapsulation
- D. Configure the TE tunnel per PE interface
- E. Configure the IP address on the VRF interface or subinterface
- F. Configure the CE to PE customer routing

Answer: A,B,E,F

Explanation:

QUESTION NO: 46

Which three layer 2 products are supported on the attachment circuit by Cisco IOS MPLS Layer 2 VPNs? (Choose three)

- A. EtherChannel
- B. Token Ring
- C. HDLC
- D. Multilink PPP
- E. Frame Relay
- F. AAL5

Answer: C,E,F

Explanation:

QUESTION NO: 47

Customer A is experiencing intermittent network performance problems. The SP NOC has determined that the "MPLS blackhole" is occurring during network convergence events. The

suspected cause of the MPLS VPN traffic issue is that the customer's IGP (OSPF) is converging faster than the LDP. What MPLS feature can be configured to eliminate the problem?

- A. Decrease the OSPF convergence timing (throttle timer)
- B. MPLS VRF protection feature
- C. MPLS fast reroute feature
- D. MPLS LDP Sync feature in OSPF
- E. Use Cisco MPLS Auto bandwidth Allocator

Answer: D

Explanation:

QUESTION NO: 48

Router#show platform hardware capacity forwarding

L3 Forwarding Resources			
FIB TCAM usage:			
	Total	Used	%Used
72 bits (IPv4, MPLS, EoM)	245760	235930	96%
144 bits (IP mcast, IPv6)	8192	420	5%
detail:			
	Protocol	Used	%Used
	IPv4	35930	96%
	MPLS	0	0%
	EoM	0	0%
	IPv6	316	4%
	IPv4 mcast	104	1%
	IPv6 mcast	0	0%
Adjacency usage:			
	Total	Used	%Used
	1048576	94037	9%

Refer to the exhibit. A service provider sees the following output indicating some problems. What are two possible causes? (Choose two)

- A. The router has not been configured with SSO
- B. The router has an inappropriately dimensioned policy feature card.
- C. The IGP timers of the router have been overturned
- D. The router has not been configured with MPLS TE automatic tunnels and fast Reroute
- E. The router has too many entries in its routing table

Answer: B,E

Explanation:

QUESTION NO: 49

When MPLS VPN services are implemented, which CE-PE routing method does not require the use of the redistribute command to advertise customer routes between sites?

- A. Static routing
- B. RIP
- C. EIGRP
- D. OSPF
- E. BGP

Answer: E

Explanation:

QUESTION NO: 50

You have configured a Layer 2 VPN circuit between two PE routers, PE1 and PE4. For some reason the VC circuit is not coming up. You checked the status and VC was down at both sides. After careful verification you have ruled out any Layer 2 VPN control plane configuration issues.

Assuring that PE devices are running cisco IOS, what two additional troubleshooting steps should help identifying the problem? (Choose two)

- A. Show mpls forwarding-table for local circuit mismatch
- B. Show mpls l2transport hw-capability to check hardware capability
- C. show mpls l2transport binding to check for MTU
- D. Show MPLS l2transport binding to check VC labels
- E. ping mpls pseudowire to check MTU mismatches in the core

Answer: C,E

Explanation:

QUESTION NO: 51

What does this Cease Notification Syslog message indicate?

May 19.12.43.45 CET: % BGP-3-NOTIFICATION: received from neighbor 10.1.2.3 6/1 (cease) 0 bytes.

- A. The port is shut down
- B. The connection was rejected
- C. BGP became disabled
- D. The maximum number of prefixes was reached

Answer: D

Explanation:

QUESTION NO: 52

You want to collect net flow-related information from your core links. You also want label switch traffic to be included in the information collected by NetFlow. You have MPLS, MPLS VPN, and protected TE tunnel in your core. How many labels should NetFlow analyze to collect information for all types of MPLS applications used in your core network?

- A. Two labels
- B. Three labels
- C. four labels
- D. five labels

Answer: C

Explanation:

QUESTION NO: 53

When MPLS layer 3 VPN services are implemented, what is the function of the VRF table?

- A. The VRF table determines the pseudowire virtual circuits used to forward labeled packets
- B. Multiple VRF instances are used to isolate potentially overlapping routing and forwarding information for VPNs.
- C. VRF tables are used to translate IP routing prefixes
- D. OSPF and IS-IS routing prefixes are MAPPED to their corresponding MPLS labeled prefixes in the VRF table. The prefixes are used by the PE to map forwarded packets between MPLS and NON-MPLS circuits

Answer: B

Explanation:

QUESTION NO: 54

Which cisco IOS command should be used to verify that frames are being forwarded on a VC by an egress PE?

- A. Show MPLS forwarding-tables
- B. Show MPLS interface forwarding
- C. Show 12 VPN counters forwarding session
- D. Show 12 VPN counters forwarding statistics

Answer: A

Explanation:

QUESTION NO: 55

Your company has decided to deploy a MPLS network with VPNs. The customer that will be using the network is demanding service level agreements with your company. How can you monitor the SLAs for violations?

- A. RMON
- B. Event management system
- C. Asset management system
- D. CNR
- E. IP SLA

Answer: E

Explanation:

QUESTION NO: 56

Which command should be used to verify that IS-IS has been properly configured to support MPLS TE?

- A. Show mpls isis traffic-eng advertisements
- B. Show isis mpls traffic-eng adjacency-log
- C. Show isis mpls traffic-eng advertisements
- D. Show isis mpls traffic-eng tunnel
- E. Show mpls isis traffic-eng adjacency-log

Answer: C

Explanation:

QUESTION NO: 57

Which statement about LSP ping is true?

- A. LSP ping access ICMP packets
- B. A successful LSP ping ensures that the echo-reply are labeled
- C. If an IP ping exceeds, than an LSP ping succeed if MPLS is enabled along the path
- D. A successful LSP ping validates reachability to the destination along the MPLS data plane

Answer: D

Explanation:

QUESTION NO: 58

What does the auto-Tunnel template interface configure?

- A. MPLS TE tunnel mesh groups
- B. Virtual private LAN services
- C. MPLS VPN VRF instances
- D. MP-BGP peering
- E. LDP tunnels
- F. MPLS TE tunnel dynamic route

Answer: A

Explanation:

QUESTION NO: 59

You have been asked to compare parameters of layer 2 MPLS vpn services at two remote ends using the show mpls l2 binding command.

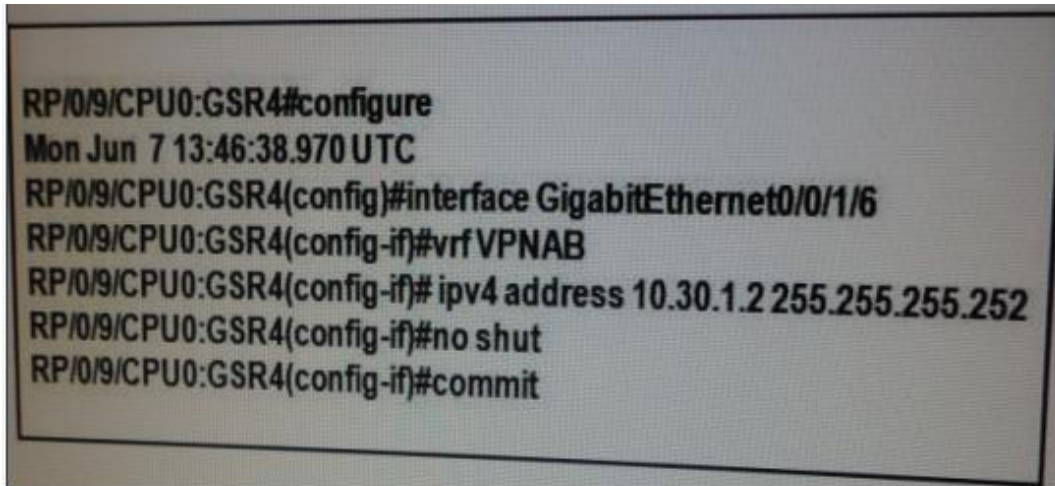
What three items of relevant information should you use to ensure that layer two VPN parameters are configured correctly? (Choose three)

- A. Local and remote labels
- B. Encapsulation
- C. VC type
- D. VCCV
- E. VC ID
- F. MTU

Answer: A,E,F

Explanation:

QUESTION NO: 60

A screenshot of a Cisco IOS XR terminal session. The text displayed is as follows:
RP/0/9/CPU0:GSR4#configure
Mon Jun 7 13:46:38.970 UTC
RP/0/9/CPU0:GSR4(config)#interface GigabitEthernet0/0/1/6
RP/0/9/CPU0:GSR4(config-if)#vrf VPNAB
RP/0/9/CPU0:GSR4(config-if)#ipv4 address 10.30.1.2 255.255.255.252
RP/0/9/CPU0:GSR4(config-if)#no shut
RP/0/9/CPU0:GSR4(config-if)#commit

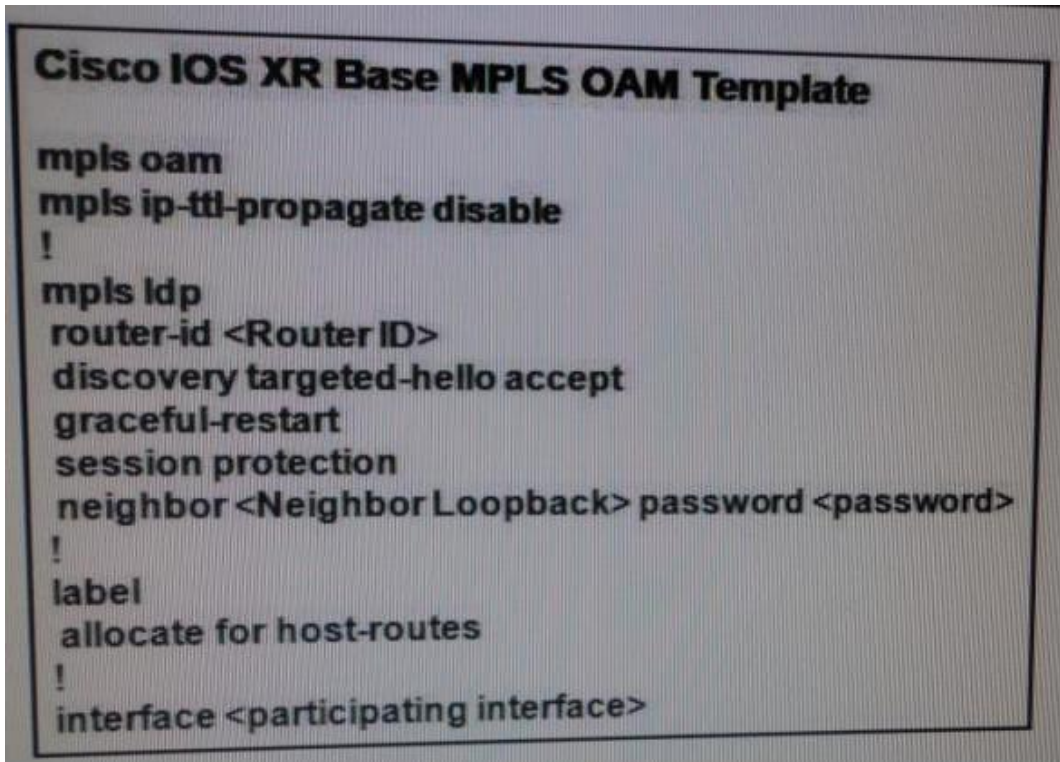
Refer to the exhibit while configuring the customer access link is cisco IOS XR software. What must be done when assigning an interface to the VRF?

- A.** Configure the neighbor LDP before accessing the interface to the VRF
- B.** Configure the IP address to the interface before assigning the interface to the VRF
- C.** Remove the customer access link before assigning the interface to the VRF
- D.** Remove any address configuring before assigning the interface to the VRF

Answer: D

Explanation:

QUESTION NO: 61



Refer to the exhibit. With this template configured and applied to cisco XR 12000series Router, could a CPE device use the traceroute command to reveal the core network?

- A. Yes, traceroute is not disabled.
- B. Yes, discovery targeted-hello accept allows it
- C. No, it is disabled with mpls ip-ttl-propagate disable
- D. No, session protection stops traceroute
- E. Only if traceroute mpls was used
- F. Only if traceroute 6 was used

Answer: C

Explanation:

QUESTION NO: 62

What function does the penultimate LSR perform to help optimize MPLS?

- A. Pops the label before the packet reaches the PE router
- B. Restore the original ingress settings to the EXP field
- C. Ensure that the LSP is Taking the shortest path
- D. Verifies that there are no loops in the path

Answer: A

Explanation:

QUESTION NO: 63

```
!
router eigrp 10
network 192.168.1.0
no auto-summary
!
address-family ipv4 vrf Customer_1
redistribute bgp 65001 metric 10000 100 255 1 1500
network 150.1.0.0
autonomous-system 20
exit-address-family
!
router bgp 65001
neighbor 192.168.1.33 remote-as 65001
neighbor 192.168.1.33 update-source Loopback0
!
address-family vpnv4
neighbor 192.168.1.33 activate
neighbor 192.168.1.33 next-hop-self
neighbor 192.168.1.33 send-community both
exit-address-family
!
address-family ipv4 vrf Customer_1
redistribute eigrp 20 metric 10
exit-address-family
```

Refer to the exhibit. The partial show run command output is from a PE router that provides MPLS VPN services to customer_1. What is the routing protocol and AS number used between the PE router and customer_1 CE router?

- A. BGP AS 65001
- B. Multiprotocol BGP
- C. EIGRP AS 10
- D. EIGRP AS 20

Answer: D

Explanation:

QUESTION NO: 64

```
Oct 11 06:33:02: %CLNS-4-LSPCKSUM: ISIS: Bad L2 LSP 0820.8200.7012.00-01 checksum E4A8 received on POS1/0/0, length 1476, should be CF62, bad count 2182
Oct 11 06:36:28: %CLNS-4-LSPCKSUM: ISIS: Bad L2 LSP 0820.8200.7020.00-01 checksum A665 received on POS1/0/0, length 1472, should be C747, bad count 2184
```


Refer to the exhibit. You are troubleshooting a connectivity problem between a CE and PE routers in your network running IS-IS. You login to one of the routers and execute a show log command and see the messages in the exhibit in the log. What is the problem associated with these messages?

- A. Protocol mismatch
- B. Mismatch area ID
- C. Mismatch MTU size
- D. Mismatch of features
- E. Wrong ip address applied

Answer: C

Explanation:

QUESTION NO: 65

What minimum commands are necessary to initiate MPLS on a cisco IOS router?

- A. In global configuration mode, execute the mpls ip command
- B. In global configuration mode, execute the mpls ip command on every participating interface.
- C. In global configuration mode, execute the mpls ip command.
Interface configuration mode executes the mpls ip command on every participating interface.
- D. In global configuration mode, execute the mpls ip command in
Mpls configuration mode specifies each participating interface using the interface interface command.

Answer: C

Explanation:

QUESTION NO: 66

Directed LDP sessions are used by MPLS to exchange labels with nonadjacent routers. How are directed LDP sessions different from normal LDP sessions?

- A. The hello packets are addressed with a statically configured destination address.
- B. The ":0" LDP identifier is added to the destination IP address to indicate the label space.
- C. Directed LDP hello messages are sent to the multicast address 224.0.0.2.
- D. TCP and UDP well-known port 646 is used for directed LDP sessions only.

Answer: A

Explanation:

QUESTION NO: 67

Which MPLS Layer 2 VPN service allows transparent bridging of VLAN trunks over MPLS?

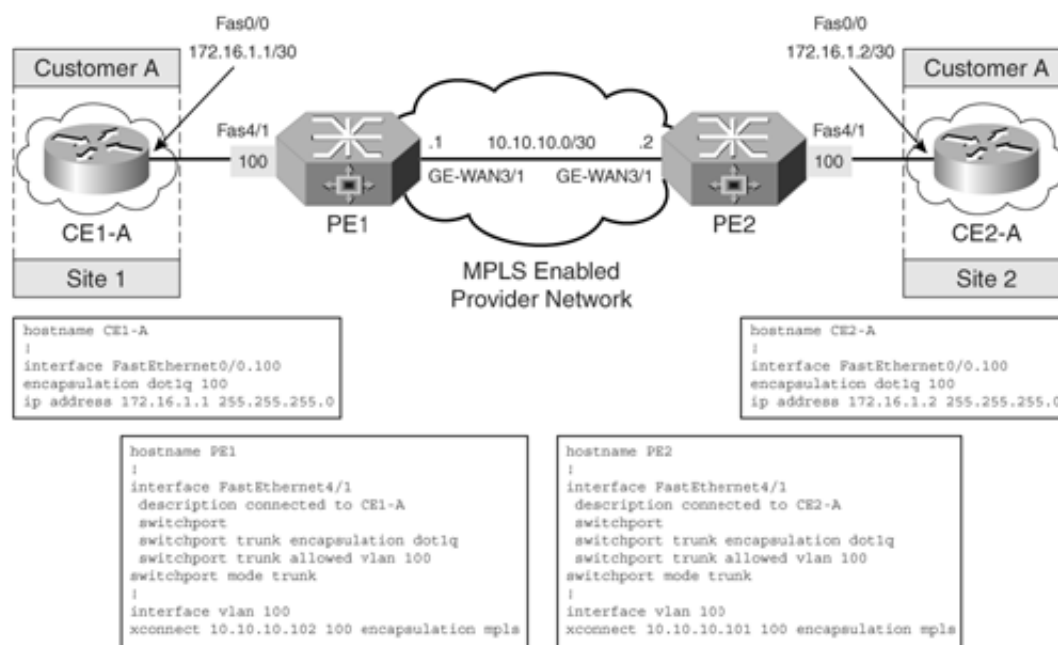
- A. Ethernet over MPLS Port Mode
- B. Ethernet over MPLS VLAN Mode
- C. MPLS Bridging
- D. MPLS VPLS

Answer: A

Explanation:

QUESTION NO: 68

Refer to the exhibit.



What Cisco IOS MPLS Layer 2 service is configured?

- A. Ethernet over MPLS Port Mode

- B. Ethernet over MPLS VLAN Mode
- C. MPLS Bridging
- D. MPLS VPLS

Answer: B

Explanation:

QUESTION NO: 69

If you are configuring MP-BGP routing for MPLS Layer 3 VPNs, when should the redistribute connected command be used?

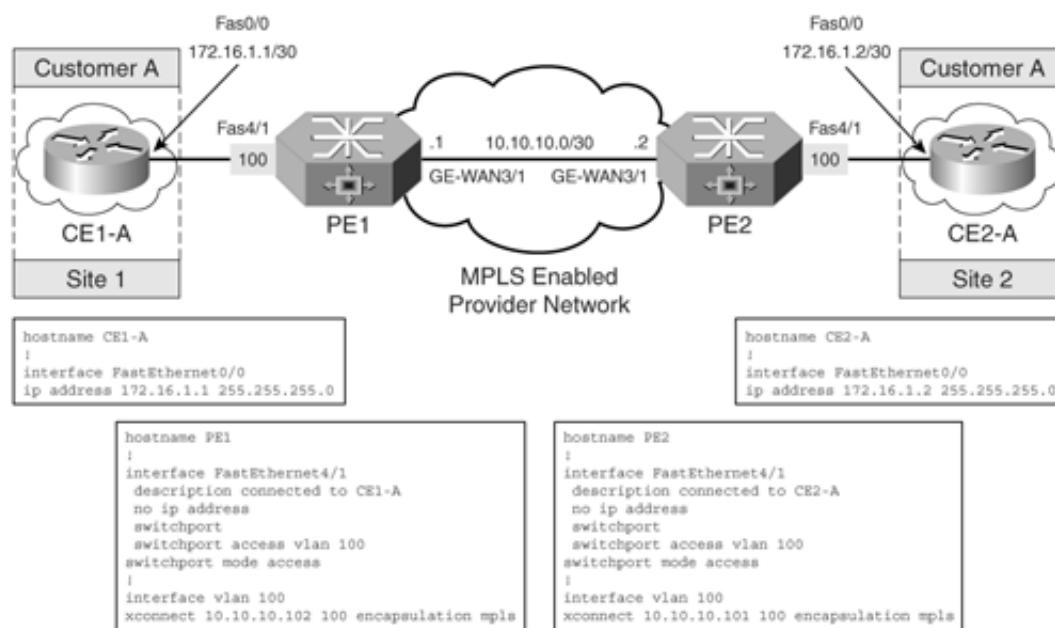
- A. when the customer requires a static default route to be advertised
- B. when the PE is supporting more than one VRF instance for the same customer CE
- C. when the customer does not have CE routers and the PE is the default gateway
- D. every time the MP-BGP redistribution function is enabled

Answer: C

Explanation:

QUESTION NO: 70

Refer to the exhibit.



What MPLS Layer 2 service is being enabled?

- A. Ethernet over MPLS Port Mode
- B. Ethernet over MPLS VLAN Mode
- C. MPLS Bridging
- D. MPLS VPLS

Answer: A

Explanation:

QUESTION NO: 71

When configuring MPLS Layer 2 VPLS, why is the Virtual Forwarding Instance needed?

- A. The VFI is used to create a full mesh of pseudowire (VC) circuits.
- B. The VFI learns the destination MAC addresses.
- C. The VFI lists the PE endpoints for Auto Discovery.
- D. The VFI defines the cross-connect group.

Answer: B

Explanation:

QUESTION NO: 72

You have been asked to verify if the OSPF protocol is enabled for Layer 3 MPLS VPN service on the link between the PE and CE.

Assume the PE is running on Cisco IOS XR software. Which command should be used to verify that the OSPF protocol is enabled for Layer 3 MPLS VPN service on a link between a PE router and a CE router?

- A. show ip ospf interface brief
- B. show vrf VRFNAME detail
- C. show ospf vrf VRFNAME interface brief
- D. show ospf interface brief
- E. show ip protocols

Answer: C

Explanation:

QUESTION NO: 73

Which command is most likely to be useful when you troubleshoot a unidirectional connectivity problem on a VC?

- A. show mpls l2transport vc
- B. show mpls l2transport vc detail
- C. show mpls l2transport checkpoint
- D. show mpls l2transport summary

Answer: B

Explanation:

QUESTION NO: 74

What Cisco IOS command should be used to obtain tunnel statistics for dynamically created primary tunnels?

- A. show mpls traffic-eng tunnels auto-bw brief
- B. show mpls traffic-eng tunnels
- C. show mpls traffic-eng tunnels statistics
- D. show mpls traffic-eng topology

Answer: B

Explanation:

QUESTION NO: 75

You are troubleshooting a Layer 3 VPN issue between three sites (site A, site B, and site C) connected in a full mesh topology. You use the show bgp vrf VRFNAME command on the PE router (Cisco IOS XR based devices) connected to site B and you find out that the only routes present are from the two remote sites (site A and site C).

What are three possible causes for the routing issue? (Choose three.)

- A. Network commands are missing under MP-BGP in remote PE routers.
- B. The redistribute command is missing under MP-BGP in remote PE routers.
- C. Network commands are missing under MP-BGP in the local PE router.
- D. The BGP neighbor prefix limit is configured on the local PE router.
- E. The BGP neighbor prefix limit is configured on remote PE routers.
- F. The redistribute command is missing under MP-BGP in the local PE router.

Answer: C,D,F

Explanation:

QUESTION NO: 76

You are using VPLS service to transport video across different points in your network. VPLS circuits are discovered automatically using the MP-BGP. A new PE device was added to the network, but VPLS circuits were not discovered by the new PE device.

Assuming that PE devices are running Cisco IOS, what two important things must you look for to identify the issue? (Choose two.)

- A. The VPNv4 VPLS address family is enabled under BGP.
- B. The Layer 2 VPN VPLS address family is enabled under BGP.
- C. The IPV4 address family is enabled under BGP.
- D. The extended BGP communities are enabled under BGP.
- E. The maximum route limit is configured under BGP.

Answer: B,D

Explanation:

QUESTION NO: 77

The SP NOC has determined that a significant number of broadcast packets are being received from customer A, which could be a DoS attack. Customer A is using the MPLS Layer 2 VPLS services. The excessive broadcast traffic is causing degraded performance and over-utilizing the access links. Which of the Cisco IOS XR MPLS Layer 2 configuration parameters can be used to mitigate the problem?

- A. the storm-control parameter under the l2vpn bridge group
- B. the storm-control parameter under the l2vpn xconnect group
- C. the limit-broadcast parameter under the l2vpn bridge group
- D. the limit-broadcast parameter under the l2vpn xconnect

Answer: A

Explanation:

QUESTION NO: 78

In a Service Provider Layer 3 MPLS VPN implementation, what is the minimum number of routing or distribution protocols running on the PE routers?

- A. three
- B. four
- C. five
- D. six

Answer: A

Explanation:

QUESTION NO: 79

How does a Label Switch Router recognize the label at the bottom of a label stack?

- A. The bottom-of-stack bit is set.
- B. The TTL field contains the number 1.
- C. The EXP field contains the number 1.
- D. The last bit in the label field is cleared in the bottom label.

Answer: A

Explanation:

QUESTION NO: 80

Which protocol must be enabled to successfully deploy Layer 3 MPLS VPNs?

- A. OSPF
- B. IS-IS
- C. EIGRP
- D. MP-BGP
- E. RSVP

Answer: D

Explanation:

QUESTION NO: 81

What function does PCALC perform for MPLS TE?

- A. Properly load balances data over MPLS TE tunnels.
- B. Calculates the shortest path based on more than cost.
- C. Tracks the performance of core routers for MPLS TE tunnels.
- D. It is the calculation used to determine if a router has the resources to accept an RSVP request.

Answer: B

Explanation:

QUESTION NO: 82

What does the MP-BGP AFI identifier indicate?

- A. which protocol an update is for
- B. whether updates are for MPLS VPN or MPLS TE
- C. whether or not route targets are present in the update
- D. which extended community attribute is being carried
- E. which VRF information is being carried in the update

Answer: A

Explanation:

QUESTION NO: 83

Refer to the exhibit.

Exhibit Missing

Given this partial configuration, which statement is true?

- A. The 209.165.200.230 router is a P router.
- B. The 209.165.200.230 router is a CE router.
- C. The 209.165.200.230 router is a PE router.
- D. The 209.165.200.230 router is a customer router behind the CE router.

Answer: C

Explanation:

QUESTION NO: 84

Refer to the exhibit. The command was executed on a Cisco ASR 9000 Series Router. Which statement is true?

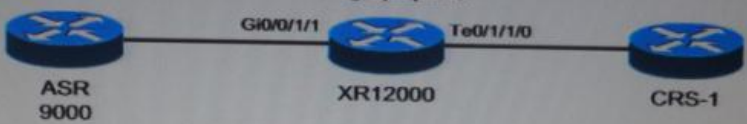
- A. The tunnel, tunnel-te1, has been configured to be used by the IGP.
- B. A maximum of 1 kb/s of traffic can pass through tunnel-te1.
- C. Traffic on tunnel-te1 will pass through two other routers before reaching the tail-end router.
- D. When an echo request is sent through the tunnel-te1, the echo-reply will come back through tunnel-te1.

Answer: A

Explanation:

QUESTION NO: 85

Refer to the exhibit.



```

debug mpls packet
ASR 9000
XR12000
CRS-1
RP/0/0/CPU0:XR12000# debug mpls packet
LC/0/0/CPU0:May 27 10:44:45.011 : netio[212]: mpls_switch-2: received an mpls packet on GigabitEthernet0_0_1_1:
proto=0, direction = ingress
LC/0/0/CPU0:May 27 10:44:45.011 : netio[212]: Flag set to 0x00000001
LC/0/0/CPU0:May 27 10:44:45.012 : netio[212]: mpls_get_payload_start: ipv4: src=0xa646402, dst=0x7f000001
LC/0/0/CPU0:May 27 10:44:45.013 : netio[212]: mpls_get_payload_start: src. port = 0xdaf, dst. port = 0xdaf, protocol =
0x11
LC/0/0/CPU0:May 27 10:44:45.013 : netio[212]: mpls_switch: GigabitEthernet0_0_1_1, mpls eos 1, ttl 1, len 132, inlabel
16017, tbi_id=0xe0000000, vrf_id=0x60000000
LC/0/0/CPU0:May 27 10:44:45.014 : netio[212]: [mpls/netio/src/mpls_netio_switch.c:1133] t3_len:[0x00000078]
readoff[0x0000000C]
LC/0/0/CPU0:May 27 10:44:45.014 : netio[212]: Flag set to 0x00000001
LC/0/0/CPU0:May 27 10:44:45.015 : netio[212]: CW 0x46000078
LC/0/0/CPU0:May 27 10:44:45.015 : netio[212]: CW NOT found
LC/0/0/CPU0:May 27 10:44:45.016 : netio[212]: iphdr_len [24]
LC/0/0/CPU0:May 27 10:44:45.016 : netio[212]: mpls_netio_is_ispv_pkt: Packet is a LSP ping packet. UDP dest
port=3503
LC/0/0/CPU0:May 27 10:44:45.017 : netio[212]: [mpls/netio/src/mpls_netio_switch.c:1307] network_start pointing at:
0x00000046
LC/0/0/CPU0:May 27 10:44:45.017 : netio[212]: First 4 network bytes: 0x46000078
LC/0/0/CPU0:May 27 10:44:45.017 : netio[212]: mpls_switch: MPLS TTL expired on LSP ping packet. Send to ipv4
netio for processing
LC/0/0/CPU0:May 27 10:44:45.030 : netio[212]: mpls_switch-2: received an mpls packet on GigabitEthernet0_0_1_1:
proto=0, direction = ingress
LC/0/0/CPU0:May 27 10:44:45.030 : netio[212]: Flag set to 0x00000001
LC/0/0/CPU0:May 27 10:44:45.031 : netio[212]: mpls_get_payload_start: ipv4: src=0xa646402, dst=0x7f000001
LC/0/0/CPU0:May 27 10:44:45.031 : netio[212]: mpls_get_payload_start: src. port = 0xdaf, dst. port = 0xdaf, protocol =
0x11
LC/0/0/CPU0:May 27 10:44:45.032 : netio[212]: mpls_switch: GigabitEthernet0_0_1_1, mpls eos 1, ttl 2, len 132, inlabel
16017, tbi_id=0xe0000000, vrf_id=0x60000000
LC/0/0/CPU0:May 27 10:44:45.032 : netio[212]: mpls_switch: DISPOSITION (PHP) to ip: prop. ttl = 1, ttl=1

```

The output from the debug command is from the Cisco XR12000 Series Router. Which statement is true?

- A. The output is the result of a single iteration of mpls ping executed on the Cisco ASR 9000 Series Router.
- B. The output is the result of a single iteration of mpls traceroute executed on the ASR 9000.
- C. The packet did not reach its destination because the control word was not found.
- D. The output is the result of a single iteration of mpls ping executed on the Cisco CRS-1 Carrier Routing System.

Answer: B

Explanation:

QUESTION NO: 86

Refer to the exhibit.

MPLS header: 0x00 0x05 0x4B 0x7E

class-map match-any high-priority

match mpls experimental 5

class-map match-any app-data

match mpls experimental 1 3 4

class-map match-any internet

Based on the MPLS header, what class of traffic is matched against the “match mpls experimental 0” packet?

- A.** class high-priority
- B.** class app-data
- C.** class internet
- D.** class default

Answer: A

Explanation:

QUESTION NO: 87

Refer to the exhibit.

*****Exhibit is missing*****

An operator is about to enter this configuration into a PE router. What is the problem with this configuration?

- A.** It uses the wrong command for starting immediately.
- B.** The prefix must be specified in dotted decimal notation.
- C.** It does not specify the test protocol first.
- D.** The schedule line should begin "schedule ipsla".

Answer: C

Explanation:

QUESTION NO: 88

Your company has decided to deploy a MPLS network with VPNs. The customer that will be using the network is demanding service level agreements with your company. What OSS system should

violations be sent to?

- A. RMON
- B. event management system
- C. asset management system
- D. CNR
- E. IP SLA

Answer: B

Explanation:

QUESTION NO: 89

Which protocol must be enabled to successfully deploy MPLS TE?

- A. OSPF
- B. IS-IS
- C. EIGRP
- D. BGPv4
- E. MP-BGP
- F. RSVP

Answer: F